

Applied Linear Algebra

Standard Deviation

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For any vector $\mathbf{x} \in \mathbb{R}^n$, the *mean* of its entries is

$$\text{avg}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n x_i$$

and the associated *de-meaned* vector is

$$\bar{\mathbf{x}} = \mathbf{x} - \text{avg}(\mathbf{x})\mathbf{1}_n$$

Notice $\text{avg}(\bar{\mathbf{x}}) = 0$, that is, the mean value of the entries of $\bar{\mathbf{x}}$ is zero.

Let $\mathbf{x} = (10, -2, 8, 23, 6, 75)$.

Then $\text{avg}(\mathbf{x}) = 20$ and the de-meaned vector is

$$\begin{aligned}\bar{\mathbf{x}} &= (10, -2, 8, 23, 6, 75) - (20, 20, 20, 20, 20, 20) \\ &= (10 - 20, -2 - 20, 8 - 20, 23 - 20, 6 - 20, 75 - 20) \\ &= (-10, -22, -12, 3, -14, 55)\end{aligned}$$

The mean of the entries of $\bar{\mathbf{x}}$ is zero.

What is $\bar{x}$ if all entries of $\mathbf{x}$ are equal?

Norm of a De-meaned Vector

Recall $\mathbf{x} = (10, -2, 8, 23, 6, 75)$ and $\text{avg}(\mathbf{x}) = 20$.

$$\|\mathbf{x}\| = \sqrt{100 + 4 + 64 + 529 + 36 + 5625} = \sqrt{6358} \approx 79.7371.$$

$$\begin{aligned}\bar{\mathbf{x}} &= (10, -2, 8, 23, 6, 75) - (20, 20, 20, 20, 20, 20) \\ &= (-10, -22, -12, 3, -14, 55)\end{aligned}$$

Norm of the de-meaned vector $\bar{\mathbf{x}}$ is

$$\begin{aligned}\|\bar{\mathbf{x}}\| &= \sqrt{(-10)^2 + (-22)^2 + (-12)^2 + 3^2 + (-14)^2 + 55^2} \\ &= \sqrt{10^2 + 22^2 + 12^2 + 3^2 + 14^2 + 55^2} \\ &= \sqrt{100 + 484 + 144 + 9 + 196 + 3025} \\ &= \sqrt{3958} \\ &\approx 62.9126\end{aligned}$$

The *standard deviation* of an n -vector $\mathbf{x}$ is defined as the RMS value of the de-meaned vector $\mathbf{x} - \text{avg}(\mathbf{x})\mathbf{1}_n$:

$$\begin{aligned}\text{std}(\mathbf{x}) &= \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \text{avg}(\mathbf{x}))^2} \\ &= \sqrt{\frac{(x_1 - \text{avg}(\mathbf{x}))^2 + \cdots + (x_n - \text{avg}(\mathbf{x}))^2}{n}} \\ &= \frac{\|\bar{\mathbf{x}}\|}{\sqrt{n}} \\ &= \frac{\|\mathbf{x} - ((\mathbf{1}_n^\top \mathbf{x})/n) \mathbf{1}_n\|}{\sqrt{n}}\end{aligned}$$

Consider our example

$$\mathbf{x} = (10, -2, 8, 23, 6, 75)$$

$$n = 6$$

$$\text{avg}(\mathbf{x}) = 20$$

$$\bar{\mathbf{x}} = (-10, -22, -12, 3, -14, 55)$$

$$\|\bar{\mathbf{x}}\| \approx 62.9126$$

Now we can compute the standard deviation of $\mathbf{x}$:

$$\begin{aligned}\text{std}(\mathbf{x}) &= \|\bar{\mathbf{x}}\|/\sqrt{n} \\ &= 62.9126/\sqrt{6} \\ &\approx 25.686\end{aligned}$$

Standard Deviation - A “Good” Sample

Unit 1

Let $x = (10, 2, 9, 18, 6, 15, 12, 7, 11)$

$$\text{avg}(x) = 10$$

$$\begin{aligned}\|x\| &= \sqrt{100 + 4 + 81 + 324 + 36 + 225 + 144 + 49 + 121} \\ &= \sqrt{1084} \approx 32.9242\end{aligned}$$

$$\bar{x} = (0, -8, -1, 8, -4, 5, 2, -3, 1)$$

$$\begin{aligned}\|\bar{x}\| &= \sqrt{0 + 64 + 1 + 64 + 16 + 25 + 4 + 9 + 1} \\ &= \sqrt{184} \approx 13.5647\end{aligned}$$

$$\text{std}(x) = \frac{13.5647}{\sqrt{9}} = 4.522$$

- The standard deviation of a vector is **zero** only when all its entries are equal.
- The standard deviation of a vector is *small* when the entries of the vector are nearly the same.

The standard deviation of a vector $\mathbf{x}$ is useful when the data set does not contain outliers, as it provides a more accurate measure of the spread of the data.

Assume a vector $\mathbf{x}$ and any number a .

- Adding a constant to every entry of a vector does not change its standard deviation.

$$\text{std}(\mathbf{x} + a \mathbf{1}_n) = \text{std}(\mathbf{x})$$

- Multiplying a vector by a scalar multiplies the standard deviation by the absolute value of the scalar.

$$\text{std}(a \mathbf{x}) = |a| \text{std}(\mathbf{x})$$

The *standardization* of a vector $\mathbf{x}$ is the vector

$$\mathbf{z} = \frac{\mathbf{x} - \text{avg}(\mathbf{x})\mathbf{1}_n}{\text{std}(\mathbf{x})}$$

$\mathbf{z}$ is the *standardized* version of $\mathbf{x}$, and has the following properties:

- $\text{avg}(\mathbf{z}) = 0$
- $\text{std}(\mathbf{z}) = 1$

The entries of $\mathbf{z}$ are called the *z-scores* associated with the original entries of $\mathbf{x}$.

Consider the following 8-vector that has a mean of 0 and a standard deviation of 1:

$$\mathbf{x} = (-0.46, -0.51, -0.29, 0.18, 1.22, -0.79, -1.27, 1.92)$$

$$\sum_{i=1}^8 x_i = 0, \quad \text{avg}(\mathbf{x}) = 0$$

$$\sum_{i=1}^8 x_i^2 = 8, \quad \|\mathbf{x}\| = \sqrt{8}$$

$$\begin{aligned} \text{std}(\mathbf{x}) &= \|\mathbf{x}\|/\sqrt{8} \\ &= 1 \end{aligned}$$

Mean Absolute Deviation

The Mean Absolute Deviation (MAD) of a vector $\mathbf{x}$ is

$$\text{MAD}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n |x_i - \text{avg}(\mathbf{x})|$$

$$\begin{aligned}\text{MAD}(\mathbf{x}) &= \frac{0.46 + 0.51 + 0.29 + 0.18 + 1.22 + 0.79 + 1.27 + 1.92}{8} \\ &= \frac{6.64}{8} \\ &\approx 0.83\end{aligned}$$

Notice that the MAD is smaller than **Std**, and in this case, MAD looks like a better measure of the typical deviation compared to **Std**.